SY B.Tech. – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PRACTICAL ASSIGNMENT TEMPLATE

1. Student Details

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Division	A
Batch	A1
Roll No.	202501110025
Date of Submission	01-11-2026

2. Practical Details

Practical No.	6
Practical Title	IMPLEMENT DIFFERENT TYPES OF INHERITANCE
Date Performed	01-11-2026
Faculty Name	Dr. Khushal Khairnar

3. Problem Statement

Write a C++ program to demonstrate four types of inheritance: single, multilevel, multiple, and hierarchical inheritance. Create suitable base and derived classes and call their member functions to show how properties and behaviours are inherited.

4. Objectives

1. Understand inheritance in C++. 2. Implement single and multilevel inheritance. 3. Demonstrate multiple and hierarchical inheritance. 4. Understand code reusability and relationships between base and derived classes.

5. Expected Learning Outcomes

1. Define base and derived classes. 2. Explain single and multilevel inheritance. 3. Implement multiple inheritance using more than one base class. 4. Implement hierarchical inheritance with multiple derived classes.

6. Theory / Concept

Inheritance: An OOP feature in which a derived class acquires accessible members of a base class.

Single: One derived class inherits from one base class.

Multilevel: A class is derived from another derived class, forming a chain.

Multiple: One derived class inherits from two or more base classes.

Hierarchical: Two or more derived classes inherit from the same base class.

7. Algorithm / Pseudocode

START
1. Define base and derived classes for each inheritance type.
2. Derive Student from Person for single inheritance.
3. Derive Mammal from Animal and Dog from Mammal for multilevel inheritance.
4. Derive AllRounder from Academic and Sports for multiple inheritance.
5. Derive Circle and Rectangle from Shape for hierarchical inheritance.
6. Create objects and call inherited and own member functions.
7. Display results.
STOP

8. Technology / Tools / Libraries Used

Programming Language	C++
IDE / Platform	Dev C++
Libraries / Packages	iostream

9. Implementation / GitHub Repository

GitHub Repository URL	https://github.com/pranav-4797/OOP-Practical-Assignment
Repository Name	OOP-Practical-Assignment
Practical Folder	Practical 6
Branch Used	main
Commit ID / Latest Commit	Not provided

Repository Contents Used

Source file: Practical_6_Inheritance.cpp. Add it to the Practical 6 folder in the repository if required.

10. Test Cases / Input and Output

Test	Type	Expected Behaviour	Status
1	Single	Student calls Person and Student functions	Pass (sample)
2	Multilevel	Dog calls Animal, Mammal and Dog functions	Pass (sample)
3	Multiple	AllRounder calls Academic and Sports functions	Pass (sample)
4	Hierarchical	Circle and Rectangle call Shape functions	Pass (sample)

11. Results / Observations

The program demonstrates single, multilevel, multiple, and hierarchical inheritance. The output shows derived classes using accessible member functions inherited from their base classes.

12. Source Code

```
#include <iostream>
using namespace std;

// Single inheritance
class Person {
public:
void showPerson() { cout << "Person class: Basic details" << endl; }
};
class Student : public Person {
public:
void showStudent() { cout << "Student class: Single inheritance" << endl; }
};

// Multilevel inheritance
class Animal {
public:
void eat() { cout << "Animal can eat" << endl; }
};
class Mammal : public Animal {
public:
void walk() { cout << "Mammal can walk" << endl; }
};
class Dog : public Mammal {
public:
void bark() { cout << "Dog can bark" << endl; }
};
```

```
// Multiple inheritance
class Academic {
public:
void study() { cout << "Academic activities" << endl; }
};
class Sports {
public:
void play() { cout << "Sports activities" << endl; }
};
class AllRounder : public Academic, public Sports {
public:
void showAllRounder() { cout << "Multiple inheritance demonstrated" << endl; }
};

// Hierarchical inheritance
class Shape {
public:
void showShape() { cout << "Shape is the base class" << endl; }
};
class Circle : public Shape {
public:
void drawCircle() { cout << "Circle is derived from Shape" << endl; }
};
class Rectangle : public Shape {
public:
void drawRectangle() { cout << "Rectangle is derived from Shape" << endl; }
};

int main() {
cout << "SINGLE INHERITANCE" << endl;
Student s;
s.showPerson();
s.showStudent();

cout << "\nMULTILEVEL INHERITANCE" << endl;
Dog d;
d.eat();
d.walk();
d.bark();

cout << "\nMULTIPLE INHERITANCE" << endl;
AllRounder a;
a.study();
a.play();
a.showAllRounder();

cout << "\nHIERARCHICAL INHERITANCE" << endl;
Circle c;
Rectangle r;
c.showShape();
c.drawCircle();
r.showShape();
r.drawRectangle();

return 0;
}
```

Screenshots / Output

Practical 6 - Inheritance - Output

```
SINGLE INHERITANCE
Person class: Basic details
Student class: Single inheritance

MULTILEVEL INHERITANCE
Animal can eat
Mammal can walk
Dog can bark

MULTIPLE INHERITANCE
Academic activities
Sports activities
Multiple inheritance demonstrated

HIERARCHICAL INHERITANCE
Shape is the base class
Circle is derived from Shape
Shape is the base class
Rectangle is derived from Shape
```

Representative terminal-style output. Run the program in Dev-C++ to capture an actual IDE screenshot if required.

13. Conclusion

This practical demonstrates four major types of inheritance in C++. It shows how derived classes reuse accessible functionality from one or more base classes.

14. Assignment Questions

Q1. What is inheritance?
Answer: An OOP feature in which a derived class acquires accessible members of a base class.

Q2. What is single inheritance?
Answer: One derived class inherits from one base class.

Q3. What is multilevel inheritance?
Answer: A derived class becomes the base class for another class.

Q4. What is multiple inheritance?
Answer: A derived class inherits from two or more base classes.

Q5. What is hierarchical inheritance?
Answer: Multiple derived classes inherit from one common base class.

15. Student Declaration

I hereby declare that the above practical work has been completed by me and that the implementation, analysis, and answers are my own work.

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Signature	